

Final Exam - MLDM 24-26 - Optimization

Volodimir Mitarchuk Antoine Gourru

Convexity

Study the convexity or concavity of the following functions and find their optimum.

$$f_1(x, y) = x^2 + 3y^2 \quad f_2(x, y) = \ln(x) + \ln(y), \text{ with } (x > 0, y > 0) \quad f_3(x, y, z) = x^2 + y^2 - z^2$$

Optimal Gradient

$$\arg \min_x f(x) = \frac{1}{2}x_1^2 + x_1x_2 + x_2^2$$

Questions

1. Provide the gradient of f in the form (i.e. explicit the value of B) : $\nabla_x f(x) = Bx$
2. Is the function f convex ?
3. Recall that the optimal learning rate in gradient descent implies that $\nabla_x f(x^{k+1})$ and $\nabla_x f(x^k)$ are orthogonal. From this, prove that the value of the optimal learning rate at epoch k is $\frac{\|\nabla_x f(x^k)\|^2}{\|\nabla_x f(x^k)\|_B^2}$
4. Apply two steps of the gradient descent algorithm with optimal step to minimize f starting at (2,1)

PSD matrix

Let $A \in \mathbb{R}^{m \times n}$ be a real valued matrix. Prove that $A^t A$ and AA^t are PSD matrices.

The Frobenius norm?

The Frobenius norm of a matrix $A \in \mathbb{R}^{m \times n}$ is defined as:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$$

where a_{ij} are the elements of the matrix A .

Questions

Show that the Frobenius norm is indeed a matrix norm by verifying the following properties:

1. **Non-negativity:** Show that $\|A\|_F \geq 0$ for any matrix A , and $\|A\|_F = 0$ if and only if $A = 0$.
2. **Scalar Multiplication:** Show that $\|\alpha A\|_F = |\alpha| \|A\|_F$ for any scalar α and matrix A .
3. **Triangle Inequality:** Show that $\|A+B\|_F \leq \|A\|_F + \|B\|_F$ for any matrices A and B of the same dimensions.
4. Show, by stating the Jensen inequality, that for all $x \in \mathbb{R}^n$ $\|Ax\|_2 \leq \|A\|_F \|x\|_2$ where $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$.
5. **Sub-multiplicativity:** Show that $\|AB\|_F \leq \|A\|_F \|B\|_F$ for any matrices A and B such that the product AB is defined. Hint: show that for all $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$

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 $C = AB$, we have $C = [C_1, \dots, C_p]$
where $C_i \in \mathbb{R}^m$, $C_i = AB_i$

$$ax_1 + bx_2 = -200$$

Production optimization

Context

A farm produces two types of animal feed: corn and wheat. The goal is to determine the optimal amount of each type of feed to mix in order to minimize costs while meeting the nutritional needs of the animals.

Objective

Minimize the total cost of the feed.

Data

- Cost of corn: 2 € per kilogram.
- Cost of wheat: 3 € per kilogram.
- Nutritional needs: Each animal requires 5 kg of feed per day, consisting of at least 2 kg of corn and 1 kg of wheat.
- Number of animals: 100.

Questions

1. Describe the linear optimization problem. ✓
2. Exhibit the cost vector $\mathbf{c}$, the constraint matrix $\mathbf{A}$ and the constraint vector $\mathbf{b}$. ✓
3. Propose a graphical solution. $1/2$

Minimizing raw material

We want to construct a lidless rectangular cardboard box with a volume of 4 liters, using the least amount of cardboard possible. The goal is to determine the dimensions of the box.

Let x , y , and z be the dimensions of the box, expressed in dm (since 1 liter equals 1 dm^3), with z being the height. Naturally, x , y , and z must be positive. Since the box has no lid, the surface area of cardboard used is $xy + 2xz + 2yz$. The volume is xyz , and it must equal 4. Therefore, we need to solve the following constrained optimization problem:

$$\begin{aligned} \min_{x,y,z} \quad & xy + 2xz + 2yz \\ \text{subject to} \quad & xyz = 4 \\ & x \geq 0, y \geq 0, z \geq 0 \end{aligned}$$

Questions

1. Write the Lagrangian function $\mathcal{L}(x, y, z, v)$. Q1.5 ✓
2. Compute $\nabla_{xyz} \mathcal{L}(x, y, z, v)$, and solve $\nabla_{xyz} \mathcal{L}(x, y, z, v) = 0$. ✓
3. Use the three equations obtained in the previous question¹ along with the constraint $xyz = 4$ to find the values of the variables x , y , and z . 0

¹Multiply each equation by a suitably chosen variable so that the right-hand side is the same for each equation.